HealthData@IE – setting up health data access body services in Ireland

Deliverable 8.9

PILOT REPORT
COMPLIANCE ASSESSMENT FRAMEWORK FOR DATA QUALITY

April 2026

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The Health Information and Quality Authority (HIQA) is an independent statutory body with responsibility for setting standards for all aspects of health information and monitoring compliance against those standards, as set out in Section 8(1) of the Health Act 2007. Under the Act, HIQA is also charged with evaluating the quality of the information available on health and social care and making recommendations to the Minister and the Health Service Executive (HSE) in relation to improving the health information system.

Conflicts of Interest

None reported.

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1. Background

1.1 The European Health Data Space and HealthData@IE

The European Health Data Space (EHDS) is the first regulated EU data space established under the European Data Strategy to promote secure and trustworthy data use.[1] Supported by the 2022 Data Governance Act and 2024 Data Act, the EHDS aims to ensure citizens can control their health data while enabling its secondary use for research, innovation and policy-making.[2,3] In Ireland, the HealthData@IE project, led by the Department of Health in collaboration with the Health Information and Quality Authority (HIQA), the Health Research Board (HRB) and the Central Statistics Office (CSO), supports the creation of Health Data Access Body (HDAB) services and the national infrastructure required for data access.[4] The project also focuses on delivery of important programmes of work centred on data quality enhancement, engagement and dissemination, as well as training and education for data users, data holders, HDAB services staff and members of the public.

1.2 Overview of Work Package 8: Data quality enhancement

Work Package 8 of the HealthData@IE project, coordinated by HIQA, comprises a programme of work centred on enhancing the quality of data that will be made available through the HDAB services and enabling data holders to improve data quality. The programme of work emphasises supporting data holders around the implementation of the EHDS and the establishment of HDAB services in Ireland, and ensuring data holders can meet their obligations relating to the secondary use of data. This work builds on the National Standards for Information Management [5], which were approved and mandated by the Minister for Health in 2024.

1.3 Requirements for data holders under EU legislation

Data holders who fall under the EHDS mandatory categories* (See Appendix A) will have to prepare metadata descriptions for each dataset, and keep details updated in a National Health Dataset Catalogue[6] so that data is easily discoverable by data users. The National Health Dataset Catalogue will build on existing health data catalogues developed by HIQA, (and will be delivered as part of Work Package 6 of the HealthData@IE project in 2027). Data holders will then have to assess and report data quality under the EHDS data quality and utility label criteria. This will be done through a common label system that specifies the quality and utility of datasets

* Under Article 51 of the EHDS Regulation, there are 17 categories of data that organisations are required to make data available to HDAB services for secondary uses. While most categories will have to be made available by 26 March 2029, some will have to be made available by 26 March 2031.

across Ireland and the EU, the reporting of which by data holders will be a legal requirement. The data quality and utility label is being established through an EU-funded project called QUANTUM. [7] HIQA is a member of the EU funded project QUANTUM along with 35 partner organisations across Europe. The EU will provide the technical specifications and procedural rules of reporting and validation of the data quality and utility label. Therefore, relevant data holders[†] in Ireland will need to review their data at regular intervals, as specified within the legislative framework. The reporting of a label, that reliably portrays quality, will be critical to accommodate the effective functioning of HDAB services, as this information will enable data users to assess whether data will be fit-for-purpose for the intended use.

1.4 Overview of the CAF-DQ

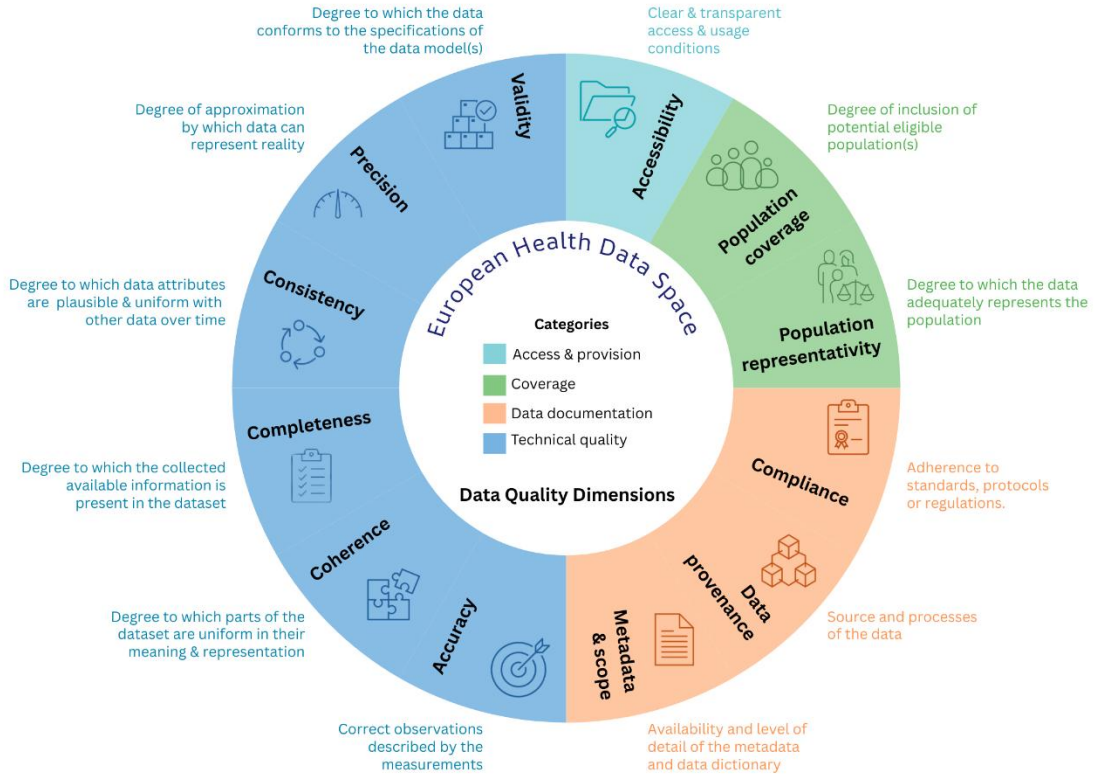
As part of the HealthData@IE project and in line with the National Standards for Information Management[7] and the associated compliance function, HIQA has developed a draft Compliance Assessment Framework – Data Quality (CAF-DQ) to outline HIQA's approach to conducting targeted reviews of data quality. This will support data holders to enhance data quality and to accurately assess and report against the label as an obligation under the EHDS Regulation.

Data holders will be obliged to assess against the EHDS data quality and utility label dimensions developed through the QUANTUM project (Figure 1).[8] Full details of the EHDS data quality dimensions, their definitions and measurements can be seen in Appendix D.

The CAF-DQ has been developed to align with the EHDS data quality and utility label and in consultation with members of the project Steering Group chaired by the Department of Health (see Appendix B for a full membership details) and the Work Package 8 Working Group (see Appendix C for full membership details).

[†] Data holders are defined as services or organisations who process personal health data that falls into any one of the categories of data for secondary use which are set out in the EHDS Regulation.

Figure 1. EHDS Data Quality Categories and Dimensions



The draft CAF-DQ was shared with pilot participants prior to the interview. The CAF-DQ is based on the EHDS data quality dimensions and will support HIQA reviewers in gathering evidence when conducting a targeted review of data quality by establishing expected practices across four levels of compliance.[‡]

The draft CAF-DQ will be revised based on the findings of the pilot along with additional engagement and the finalised version will be published in October 2027. The team in HIQA are also developing the methodology for the targeted reviews which will be published in an updated version of the *Guide to the Health Information and Quality Authority's review of information management practices in national health and social care data collections* before targeted reviews commence.

[‡] HIQA compliance levels classify services as Compliant, Substantially Compliant, Partially Compliant, or Non-Compliant with national standards.

2. Introduction

2.1 Purpose of the CAF-DQ pilot

The overall aim of the CAF-DQ pilot is to assess the feasibility of implementing the CAF-DQ, focusing specifically on the implementation of the 12 EHDS data quality dimensions with data holders in Ireland in preparation for EHDS. As the EHDS data quality dimensions and their measurements are novel, the aim was to assess if and how these dimensions could be applied in practice by data holders of varying levels of maturity.

The specific objectives of the CAF-DQ pilot are to:

- assess the feasibility of using the EHDS data quality dimensions with health data holders of different types and maturities, in an Irish context
- use this information to review current data quality process and support updates to the CAF-DQ in line with evolving requirements and practices at a European level.

2.2 Purpose of the report

The purpose of this report is to outline the key findings of the CAF-DQ pilot. These findings will be used to:

- inform and guide future iterations of the CAF-DQ and any additional resources currently being developed before they are published and utilised
- identify areas where data holders may need to further develop data quality practices to support implementation of the EU Data Quality and Utility label and compliance in line with the CAF-DQ.

3. Methodology

3.1 Participants

The CAF-DQ was piloted with three data holders from across various settings. In early 2025, HIQA conducted a survey of data holders' levels of preparedness for the establishment of HDAB services in Ireland, called the Readiness Assessment.[9] A sample of data holders involved in the Readiness Assessment were invited to participate in the CAF-DQ pilot, representing different types and maturity of data holders as identified through the Readiness Assessment findings. Two of the data holders were national data collections and one of the data holders was an eHealth system.

3.2 Methods

For the pilot, each participating data holder was invited to take part in an individual feasibility study which was conducted as a semi-structured interview. Each session lasted 2.5-3 hours. Participants received a version of the CAF-DQ and background briefing documents outlining the purpose and key themes for discussion, via email, at least two weeks in advance of these sessions. Data holders were asked to review the CAF-DQ before the session and discuss any feedback during the session. Data holders were informed that the pilot was not assessing compliance with the framework, as this would require an in-depth review of documentation and practices, but instead was assessing the feasibility of applying the CAF-DQ and EU Data Quality and Utility dimensions in practice. These findings would be then used to further develop the CAF-DQ.

The sessions were conducted in person and were guided by a member of the HIQA team, using a semi-structured topic discussion guide. The sessions were recorded and transcribed for note-taking purposes.

The topic discussion guide was developed with input from the Work Package 8 Project team and was organised into three areas:

- **Data quality in the organisation.** During part one of the session, participants were asked general questions to get an overall understanding of their data quality practices.
- **Initial observations of the data quality dimensions.** During part two of the session, organisations were given a presentation on the 12 data quality dimensions and were asked for their initial feedback on the clarity and understandability of the dimensions and their measurements.
- **Application of the data quality dimensions in the organisation.** In part three of the session, each data quality dimension was discussed in detail, with emphasis on whether this is an area currently assessed and reported on by the organisation, and if so, how this assessment and report is conducted? If not, then what were the challenges and barriers anticipated?

3.3 Analysis

A framework analysis approach was used to analyse the data. Transcripts and audio recordings from the three sessions were reviewed, and a thematic framework was drafted using the topics outlined in the discussion guide. The three transcripts were imported into NVivo to support a systematic and transparent coding process.

To ensure analytical rigour, a second iteration of this process was performed by another team member, with re-review of transcripts to identify any discrepancies.

Differences in interpretation were discussed and resolved through collaborative discussion to reach a consensus, and uncertainties were discussed with a third team member.

The final synthesis and interpretation involved a team-based approach, where key findings were explored and reviewed and each theme and subtheme were considered in the context of the whole set of interviews.

4. Key findings

4.1 Summary of key findings

Table 1. Key findings and insights from participants

Dimension[§]	Key findings
Accessibility	Participants stated that reporting on accessibility is generally straightforward. However, more guidance is needed on the scope of this dimension (including accessibility through open data) and to ensure all organisations are tracking timelines in a standardised way once a complete data request is submitted.
Population Coverage	The concept of population coverage was well understood, however currently participants do not routinely report on this dimension and there are gaps in knowledge on best practice methods to assess this, particularly for datasets that are based on the utilisation of services.
Population Representativity	Participants reported that they would find it difficult to assess this dimension but particularly for datasets driven by clinical need as representativity was viewed as difficult to measure. Participants found that for certain datasets, methodology has not been fully established which made reporting on it challenging.
Compliance	Participants noted challenges in reporting on compliance due to its broad definition and metric, and limited guidance on the scope of this dimension. For example, should this include broader information governance protocols and standards or focus specifically on data quality?

[§] Please see Appendix D for definitions and metrics on each dimension.

Dimension[§]	Key findings
Data Provenance	Participants could conceptually describe data provenance practices but noted that documentation is often performed across different organisations and departments, at present. Therefore, data holders do not currently have control or oversight of all the relevant documentation, or documentation is not in a standardised or agreed format.
Metadata Scope	Participants reported that are currently not using Health Data Catalogue Vocabulary - Application Profile (Health DCAT-AP). Most participants have a data dictionary to some degree, however there are currently no plans to align with any national data dictionary.
Accuracy	Accuracy is viewed as important and participants noted that systems are often designed to enforce valid entries, but currently they cannot feasibly validate every record or entry for accuracy. Therefore, they are unsure how they should assess and report against this dimension, other than documenting processes. Participants currently do not have the systematic or automated processes required to check the accuracy of data once it has been entered. Guidance and examples are needed to ensure data holders can align with best practice.
Coherence	Participants generally have processes in place to ensure coherence of data such as aligning to coding standards, but discussions revealed uncertainty about how to assess and report against this dimension. Guidance and examples are needed to ensure data holders can align with best practice.
Completeness	Completeness is typically well supported due to structured data entry processes that prevent incomplete records. However, participants reported that they would find it hard to improve this or even to assess and report for all variables. Guidance and examples are needed to ensure data holders can align with best practice.
Consistency	Most data holders take steps to ensure consistency, including governance processes for new variables. However, historical data presents challenges, and measuring and reporting consistency remains difficult. Guidance and examples are needed to ensure data holders can align with best practice.

Dimension[§]	Key findings
Precision	Precision was identified as one of the most challenging dimensions to interpret and report on. While steps are taken to ensure precise data, participants were unsure how to document precision or what evidence would be required. Guidance and examples are needed to ensure data holders can align with best practice.
Validity	This dimension requires use of a standardised data model which is not currently used by data holders involved. Participants expressed uncertainty about what constitutes a data model and how validity should be assessed and reported on, noting that further support to enhance skills in this area would be required. Guidance and examples are needed to ensure data holders can align with best practice.

4.2 Ireland's digital health maturity

This pilot explored the feasibility of implementing the 12 data quality dimensions developed for use across all European countries as part of the EHDS. While these dimensions provide a comprehensive and robust framework, their application in an Irish context presents feasibility challenges due to Ireland's health information system being less digitally mature than many other Member States.

Several dimensions assume levels of digital infrastructure, interoperability and system integration that are not yet in place across the Irish health information system. As a result, some dimensions are challenging for data holders in Ireland to interpret or measure. It can also be difficult to streamline or automate the process. For example, as national electronic health records (EHRs) are not currently in place in most healthcare settings, review of data quality at times would require pulling a sample of medical records to assess and report on some dimensions of data quality.

4.3 Governance and data quality

The current governance practices and structures vary across the three participant data holders. Only one data holder reported having a formal data quality framework in place. A second data holder is actively developing a data quality framework and is attempting to align it with the 12 data quality and utility dimensions. Two of the three data holders have assigned a specific role or designated responsibility for data quality.

A key finding is that most dimensions were being addressed in some form and to some extent by data holders. However, they often lack the structures, documentation and governance needed to fully meet the assessment and reporting

expectations of the Data Quality and Utility label. In all three cases, steps are already being taken to meet the intent of assessing and reporting against the dimensions. For example, putting system controls in place that require fields to be completed, which in turn support completeness and accuracy. However, these practices are not consistently measured or documented across all dimensions, making it difficult for data holders to assess and report on their performance in line with EHDS requirements. For example, a finding from participants was that they are generally able to assess and report on the processes but not the outcomes for each dimension.

Participants also highlighted that data holders can only ensure the quality of the data they receive; where source data has data quality issues, their ability to improve the quality of the data is sometimes limited. This dependency on upstream data providers affects several dimensions and underscores the need for system-wide approaches to data quality.

Participants described efforts in formalising existing practices around data quality so that these can be consistently measured, monitored and reported. They also acknowledged that strengthening governance around data quality is essential to ensure that responsibilities are clear, processes are repeatable and performance against each dimension can be reliably demonstrated over time and in line with best practice. However, they called for further support through guidance and practical examples to help organisations document their processes and outcomes in line with best practice and embed governance mechanisms that enable consistent reporting and oversight of the new data quality dimensions. They welcomed the upcoming publication of HIQA's National Guidance for Enhancing Data Quality in Health and Social Care, in line with these changes.

4.4 Interpretation and reporting of dimensions

Participants consistently reported that several of the EHDS dimensions were challenging to interpret and apply in practice. In many cases, the concepts were perceived as insufficiently defined or overly broad. This led organisations to form their own interpretations rather than work from a shared understanding. This resulted in significant variation in how dimensions were understood, assessed and explained.

Due to the perceived lack of clear definitions, practical examples and methodological guidance provided to support the assessment and reporting against the EHDS dimensions, it was challenging for organisations to determine what was expected of them. As a result, even where organisations were confident that they were meeting

the intent of a dimension, they often struggled to evidence this, due to uncertainty about the underlying requirements.

This finding highlights the need, as expressed by data holders, for clearer guidance to be developed at an EU and or national level, including practical examples to support consistent interpretation and reporting on the dimensions to ensure repeatable and standardised practice.

4.5 Access and provision

4.5.1 Accessibility

Participants generally demonstrated an ability to provide accessible data and are meeting the intent of the accessibility measurements in practice. However, none of the participants reported having a formal or documented approach for assessing or reporting on this dimension. Reporting practices in this area rely heavily on manual tracking, and participants described different methods which introduces inconsistency across organisations in terms of defining timelines (start and end points). As expressed by data holders, guidance is required on measuring the timeliness requirement to ensure data is comparable across datasets. Organisations should count from the time the initial request is submitted or from when there is a well-defined or actionable request submitted.

Participants noted that timelines are dependent on external factors. For example, complex data requests, or requests that require extensive clarification with requesters can significantly extend timelines and pose challenges to data holders in terms of resources. Participants also stated that they would find it difficult to demonstrate compliance in a systematic way and would need to further develop procedures and practices to continuously assess, report and implement improvements to enhance this dimension.

A broader issue identified is that the current definition of the accessibility dimension does not account for data made available through open-data platforms or business intelligence tools that can share data or results without needing to submit a data request. As a result, organisations capability to make data accessible in a timely manner may be underestimated, especially for those that dedicate significant resources to sharing data in open-data platforms.

4.6 Coverage

4.6.1 Population coverage

Population coverage is well understood by participants. However, these rates are not always formally documented or routinely reported. Existing gaps in population coverage are generally outside of the data holders' control and tend to arise in areas where public organisations lack insight and influence into private sector activity, making high rates of coverage at a population level difficult to achieve.

4.6.2 Population representativity

Participants found population representativity difficult to interpret and apply to their datasets. While the methodology is well-developed for sample-based datasets, it is less applicable to datasets generated through admittance, service use, or clinical need. In addition, in these cases, organisations have limited ability to influence representativity, even if a study could be conducted to assess it. Participants highlighted the need for clearer methodological guidance on how representativity should be measured and reported across different dataset types.

4.7 Data documentation

4.7.1 Compliance

The compliance dimension was found to be difficult to interpret in the context of the data holders that took part in this pilot. The measurement is perceived as vague, leading to significant variation in interpretation, from broad information governance and GDPR considerations to highly specific coding standards. While the dimensions aim to avoid being overly prescriptive, participants emphasised the need for clearer scope, definitions, and examples of good practice to support consistent assessment and comparable reporting on compliance.

4.7.2 Data provenance

All three organisations explained that they have elements of the data provenance dimension in place, including documentation, process descriptions, technical traceability and auditability. However, the origin and pathway of the data sometimes sit outside the organisation and therefore, they have limited control over documenting its provenance as well as full influence over improvements. Furthermore, gaps in traceability, documentation and governance structures mean that organisations cannot currently report on provenance in a complete or consistent way. Participants expressed a need for clearer guidance on best practice assessment and reporting requirements for this dimension. This dimension requires oversight of

the complete data lifecycle which again, emphasises the need for a system-wide approach to enhance data quality.

4.7.3 Metadata scope

Participants' current metadata maturity varies considerably across organisations. However, all participants recognise that metadata management and alignment with Health DCAT-AP are becoming increasingly important, particularly in the context of emerging EU requirements. None of the organisations involved reported that they are currently using Health DCAT-AP, and data holders expressed a strong need for guidance and structured support to meet these expectations. Some organisations maintain data dictionaries, but practices differ, and clarity is needed on how to align with the National Data Dictionary.

4.8 Technical Dimensions

4.8.1 Accuracy

All three participating organisations have processes in place to support accurate data collection, such as system controls that prevent invalid entries. Practices to enhance accuracy were described by data holders such as identifying duplicates, detecting extreme values and ad hoc checks performed by staff. However, routine assessment and reporting on this dimension is not consistently carried out by participants. Similarly, providing assurances to management, such as generating KPIs, are not consistently reported. Therefore, participants involved in the pilot cannot systematically verify the accuracy of all data, and concerns were raised about the current feasibility of assessing the accuracy of all variables, given the source of data can often be within paper-based medical records. Participants questioned what the expectation is in terms of assessing and reporting for the label in these scenarios.

4.8.2 Coherence

Participants identified that there could be an overlap between the assessment and reporting of coherence and consistency dimensions. Data holders involved approach coherence in different ways, some update coding classifications as needed, while others maintain systematic processes to track coding changes over time and manage internal reference tables. Standard coding systems are used but challenges arise as coding systems evolve. Participants highlighted the need for clearer expectations regarding documentation of both processes and outcomes for this dimension.

4.8.3 Completeness

All three data holders involved in the pilot take steps to ensure data fields are fully populated. However, there is considerable variation in how completeness is measured and reported. No standardised targets exist across participants, and resource constraints can limit the ability to address gaps. Completeness is also affected by reliance on information provided by external parties and by variation across sites and teams. Participants asked for further guidance on documentation expectations as outlined in the EHDS dimensions, specifically requesting clarity on whether completeness reporting should cover all variables or focus on a defined minimum dataset.

4.8.4 Consistency

All three data holders involved in the pilot have mechanisms to prevent uncontrolled changes and are working toward greater standardisation over time. Challenges arise when working with historic data that contains legacy inconsistencies. Consistency checks are typically embedded in day-to-day operations rather than supported by a formalised process. Participants require clearer guidance on documentation expectations and whether consistency checks should apply to all variables, or only those most prone to inconsistencies.

4.8.5 Precision

All participants emphasise that their systems are designed to support precision with drop-down menus, predefined code sets, structured fields and prompts to help ensure that data is specific and to prevent imprecise entries. However, measuring or reporting on precision is difficult and often not formally assessed or documented. Challenges also arise from inconsistent data entry practices in a clinical setting. Some participants noted how currently they have difficulty in calculating precision rates for variables. Participants requested guidance on what is expected in terms of documentation of processes and outcome for this dimension.

4.8.6 Validity

Validity in this context is defined as compliance with a data model rather than inherent correctness of the data. While organisations take steps to ensure data adheres to certain standards, they are often unclear about the data models in use and their specifications. They also noted that they currently do not have the skills and knowledge to support the development of data models. Participants highlighted the need for clearer guidance on best practice, documentation expectations, and how validity should be assessed in relation to data models.

4.9 Assessment of compliance and the CAF-DQ tool

Findings from the engagement with data holders indicate that they currently face challenges with the initial step of assessing and reporting across the 12 dimensions. All participants involved noted that they will have to undertake significant preparations before they are ready to submit a Data Quality and Utility Label. Therefore, participants stated they are not yet currently in a position to assess, evidence or report consistently across all dimensions. They also identified that they currently lack the digital maturity, processes, documentation and governance required to comprehensive enhance data quality in line with the framework. This finding is applicable despite the level of maturity and type of data holder. The organisations involved were of varying levels of maturity, as self-reported in the Readiness Assessment.**[9] Therefore, the findings of this feasibility study show that, despite maturity, none of the organisations are currently fully equipped to meet the EHDS reporting requirements against the EHDS dimensions.

Even the most mature organisation will have significant preparations ahead in terms of enhancing and documenting processes and defining Key Performance Indicators (KPIs) to align with the assessment and reporting requirements across all of the dimensions. For example, while they can put systems in place to prevent invalid entries, they could not automate or systematically report on accuracy, precision or completeness. Organisations' current data quality efforts are focused on improving the process rather than producing evidence of outputs, meaning they are doing many of the right things operationally but cannot yet demonstrate compliance in a systematic, measurable way. To develop a CAF-DQ that supports quality improvement, a clearer understanding of what is expected in terms of assessment and reporting across each dimension is needed.

Another key finding from the pilot to inform the development of the CAF-DQ is that organisations may demonstrate compliance with some, but not all, categories within a single dimension. For example, they might have the policies and processes in place but not have staff training available. Therefore, this current format will make it difficult for reviewers to make an overall assessment of compliance within and across dimensions. To address this, the format of CAF-DQ will need to be reviewed to allow more flexibility in its capture of information and judgment of compliance. A cascading or hierarchical format within the CAF-DQ may better support overall assessment of compliance across dimensions. Such an approach would support

** A readiness assessment was carried out with data holders in 2025 where they were asked to report on the maturity of their data quality with regard to the Information Management Standards for health and social care.

organisations to identify targeted areas for improvement, while giving reviewers a clearer view of overall data quality.

5. Significance of findings

Based on these findings, additional preparation is required at an EU and national level to support data holders on the implementation of the data quality dimensions in practice. This finding is reinforced by the fact that the pilot included different levels of 'maturity' among data holders and across different 'types' of data holders and all have had significant issues with the assessment and reporting of these dimensions. The findings of this pilot also strongly align with the findings of the Readiness Assessment of data holders in Ireland in relation to the establishment of a Health Data Access Body (HDAB) service.[9] The readiness assessment report identified that preparation is required at both the national level and data holders' level, to ensure Ireland is ready for the implementation of the EHDS Regulation and to enable the effective establishment of a HDAB service.

The findings of the pilot indicate that data holders are currently able to assess and report on processes rather than outcomes, across many of the dimensions. This distinction is significant, while process level activity indicates intent and emerging maturity, it does not provide the evidence required by the EHDS data quality dimensions. As data holders will be required to assess and report on their data quality practices, they will need to strengthen practices and the mechanisms for reporting on them in line with the 12 dimensions. Organisations will need to establish robust practices and processes in preparation for future assessment and reporting. In addition to submitting a Data Quality and Utility label, HIQA will be conducting targeted reviews of data quality to review practices in more detail, helping to validate evidence in the label as well as drive improvements in data quality supporting system-wide improvements.

The findings also reveal variation in governance structures and data quality practices across data holders. With only one organisation having a formal data quality framework in place and others relying on emerging or informal arrangements, the sector lacks the consistent systems and governance arrangements required to support reliable assessment and reporting, as well as oversight (and sign-off) of the label and actioning and monitoring enhancements based on the findings which will ultimately drive a cycle of continuous improvement.

Further, the findings highlight variation on how organisations have interpreted and applied the dimensions, which raises a risk of inconsistent assessment and reporting across data holders.

These findings are significant because they demonstrate a mismatch between the expectations of the EHDS data quality dimensions and the current level of maturity within the health information system in Ireland. Due to current challenges with applying the dimensions in practice, organisations will find it challenging to assess and report against the label. Further engagement at a European level will help to understand best practice, enhance interpretation and implementation of the dimensions in practice. This will also provide a better understanding of the process to ensure the CAF-DQ is fit-for-purpose.

It is acknowledged that this pilot was based on the experiences of three organisations. However, the pilot included major national data collections in Ireland that varied in type and maturity. This provides a reasonable basis for inferring that if these organisations are encountering difficulties in applying the dimensions in practice, it is likely that others are experiencing similar challenges. Consequently, we aim to develop a deeper understanding of these challenges and potential solutions, which is why further engagement is planned (see Next Steps).

Furthermore, as this is a feasibility study, we did not aim to assess and report on compliance of participating data holders. However, the findings provide a valuable indication of the challenges organisations face in achieving readiness to apply these requirements in practice, and it offers important learnings for HIQA as it develops a future compliance programme and associated guidance materials.

6. Next steps

Building on the insights gained through the pilot, the following actions will guide the continued refinement and implementation of the CAF-DQ tool, the development of methodology for the target reviews of data quality and associated guidance for data holders in Ireland:

- Provide feedback to the EU via the QUANTUM project on the findings and challenges identified through engagement with data holders in Ireland; highlighting the implications of applying the EHDS data quality dimensions in countries with lower levels of digital maturity.

- Conduct additional engagement with both European and Irish data holders to determine best practice in terms of implementing these dimensions. It will also help to gather insights into experiences of data holders in achieving data quality enhancement and to overcome challenges.

- Collaborate with data holders in Ireland to support preparations for the data quality requirements associated with the EHDS.

- Determine what changes and updates may be required to the CAF-DQ tool in line with EHDS obligations, revise and refine the tool based on these findings, and support the development of a tool that effectively measures compliance of data quality. Finalise and prepare the CAF-DQ for use by HIQA reviewers to carry out targeted reviews of data quality with data holders.

- Incorporate the resulting refinements into any supporting and guidance tools to ensure usability of the CAF-DQ tool and its alignment with the EHDS obligations and other HIQA information management tools.

- Use the findings to inform HIQA's programme of work on data quality including the development of a national guidance on data quality. This includes updating the Guidance on a Data Quality Framework and associated e-learning modules on data quality, in line with the 12 EHDS dimensions of data quality.

Appendix A – EHDS categories of data for secondary use

According to Article 51 of the European Health Data Space (EHDS), data holders shall make the following categories of electronic data available for secondary use by 2029:

1. Electronic health data from EHRs.
2. Data on factors impacting on health, including socio-economic, environmental and behavioural determinants of health. *
3. Aggregated data on healthcare needs, resources allocated to healthcare, the provision of and access to healthcare, healthcare expenditure and financing.
4. Data on pathogens that impact human health.
5. Healthcare-related administrative data, including dispensations, reimbursement claims and reimbursements.
6. Human genetic, epigenomic and genomic data. *
7. Other human molecular data such as proteomic, transcriptomic, metabolomic, lipidomic and other -omic data. *
8. Personal electronic health data automatically generated through medical devices.
9. Data from wellness applications.
10. Data on professional status, and on the specialisation and institution of health professionals involved in the treatment of a natural person.
11. Data from population-based health data registries such as public health registries.
12. Data from medical registries and mortality registries.
13. Data from clinical trials, clinical studies, clinical investigations and performance studies subject to Regulation (EU) No 536/2014, Regulation (EU) 2024/1938 of the European Parliament and of the Council, Regulation (EU) 2017/745 and Regulation (EU) 2017/746. *
14. Other health data from medical devices.
15. Data from registries for medicinal products and medical devices.
16. Data from research cohorts, questionnaires and surveys related to health, after the first publication of the related results. *
17. Health data from biobanks and associated databases.

* The inclusion of this category of data will come into effect in 2031.

Appendix B – Steering Group membership for HealthData@IE

Name	**Organisation** ***Title, Division***
Emer Doyle	Department of Health (Chair) *Principal Officer, Health Information Policy*
Rachel Flynn	Health Information and Quality Authority (HIQA) *Director, Health Information and Standards*
Barbara Foley	HIQA *Deputy Director, Health Information Quality and Assurance*
Kevin O'Carroll	HIQA (WP8) *Programme Manager, Health Information Standards*
Brendan Brady	Department of Health *Health Information Policy*
Bríd Burke	Health Research Consent Declaration Committee
Frances Burns	Northern Ireland Trusted Research Environment
Helen Conroy	Department of Health *Health Information Policy*
Eamonn Coyne	Department of Health *eHealth and Health Information Systems*
Steve Dempsey	Irish Cancer Society *Patient Representative*
Kevin Devlin	National Research Ethics Committee *HRB PPI Reviewer, Patient Representative*
Eoin Farrell	Department of Health *Health Information Policy*
Laura Flannelly	Department of Health *Statistics and Analytics Unit*
Lorna Gaffney-White	Department of Health *Health Information Policy*
Richard Greene	Health Service Executive (HSE) *Chief Clinical Information Officer*
Loretto Grogan	HSE Office of the Chief Clinical Officer *Chief Information Officer – Nursing and Midwifery*
Prof Markus Helfert	National University of Ireland, Maynooth
Noreen Hynes	Health Service Executive (HSE) *Chief Data and Analytics Office*
Ailish Kelly	Department of Health *Health Information Policy*
Kathryn Kissane	Health Service Executive (HSE) *Deputy Delivery Director, Standards & Shared Care Records*
Anthony Macken	Central Statistics Office (CSO) *Researcher Coordination Unit*
Dr Teresa Maguire	Health Research Board (HRB)

Name	**Organisation** ***Title, Division***
	Director of Research Strategy and Funding
Denise Manton	National University of Ireland, Maynooth
Elizabeth McCrohan	Department of Health *Statistics and Analytics Unit*
Elaine Murray	EIT Health (Ireland-UK) *Public Affairs*
Cáit Ní Chorcora	Department of Health *Health Information Policy*
Azul O'Flaherty	Department of Health *Health Information Policy*
Ronan O'Kelly	Department of Health *eHealth and Health Information Systems*
John O'Neill	Department of Health *Research Policy and Innovation Unit*
Róisín O'Neill	Department of Health *Research Policy and Innovation Unit*
Christopher Ryan	Department of Health *Research Services Unit*
Kerry Ryder	Health Service Executive (HSE) *Research and Development*
Niall Sinnott	Department of Health *eHealth and Health Information Systems*
Clodagh Thorne	Department of Health *Health Information Policy*
Caitriona Wray	Department of Health *Health Information Policy*

Appendix C – Working Group membership for Work Package 8

Name	**Organisation** ***Title***
Barbara Foley	Health Information and Quality Authority (HIQA, Chair) *Deputy Director, Health Information Quality and Assurance*
Maria Ryan	HIQA (WP8) *Programme Manager, Health Information Quality and Assurance*
Kevin O'Carroll	HIQA (WP8) *Programme Manager, Health Information Standards*
Anthony Macken	Central Statistics Office (CSO) *Researcher Coordination Unit*
Brendan Brady	Department of Health, Health Information Policy Unit *Data Officer & Health Literacy Policy Lead*
Cliona O'Donovan	National Office of Clinical Audit (NOCA) *Head of Data Science*
Frank Moriarty	Royal College of Surgeons in Ireland (RCSI) *Senior Lecturer*
Gabriela Avram	HSE National Patient Forum *Patient partner*
Helen Conroy	Department of Health, Health Information Policy
Jean Kelly	National Cancer Registry Ireland (NCRI) *Data Quality Manager*
Leonard Browne	University of Limerick (UL) *Senior Research Fellow (Biostatistics)*
Loretto Grogan	HSE Office of the Chief Clinical Officer *Chief Information Officer – Nursing and Midwifery*
Lorraine McNerney	Department of Public Expenditure, NDP Delivery and Reform - Office of the Government Chief Information Officer *Head of Data Governance*
Louise Cullen	HSE Health Protection Surveillance Centre *Principal Epidemiologist*
Niall Halliday	HSE Office of the Chief Clinical Officer *National Chief HSCP Clinical Information Officer*
Patricia Wilson	HSE Office of the Chief Clinical Officer *ICT Project Manager, Shared Care Record*
Sarah Craig	Health Research Board (HRB) *Director of Health Information and Evidence*

Appendix D – EHDS Data Quality Dimensions

Category	Dimension	Definition	Measurement
Access and provision	**Accessibility**	Accessibility refers to the dataset being accompanied by clear and transparent access and usage conditions.	Availability of a data access & usage policy at the time of release of the dataset. Average time from data access application to data release for a specific dataset.
Coverage	**Population coverage**	Population coverage refers to the degree to which a dataset includes the potential eligible population.	Coverage Rate (percentage of the eligible population represented in the dataset).
	Population representativity	Population representativity refers to the degree to which the data adequately represent the population in question.	How closely does the observed population represent the expected population?
Data documenta-tion	**Data provenance**	Data provenance means a description of the source of the data, including context, purpose, method and technology of data generation, documenting agents involved in the provenance of data, data validation routines, source data verification, traceability of changes, and quality control of data.	Is the source of the dataset documented? Are the processes and operations on the data documented?
	Compliance	Compliance refers to the degree to which data has attributes that adhere to standards, conventions, protocols or regulations.	Is there documentation of compliance with standards, conventions, protocols or regulations and an individual responsible for this?
	Metadata scope	Metadata scope refers to the availability, comprehensiveness, level of detail of metadata and data dictionary that help users understand the data being used.	Existence of comprehensive standardised metadata; Existence of an exhaustive data dictionary at variable level.

Technical Quality	**Validity**	Validity refers to the degree to which representations of data in a dataset conform to the specification of a data model or data models.	Availability of a conformance report for the data model.
	Accuracy	Accuracy refers to the degree to which observations correctly describe what it was designed to measure.	Is the accuracy of the dataset documented?
	Coherence	Coherence is defined as the dimension that expresses how different parts of the dataset are uniform in their representation and meaning over time, such as formats, semantics (stability of the data models), and methods.	Is coherence of the dataset documented?
	Completeness	Completeness refers to the degree to which all information that could be available is present in a particular dataset.	Is the completeness of the dataset documented?
	Consistency	Consistency refers to the degree to which data has attributes that are plausible and are uniform with other data and over time.	Is the consistency of the dataset documented?
	Precision	Precision refers to the degree of approximation by which data can represent reality.	Is the precision of the dataset documented?

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